

Lab 4: Spatial Analysis of Crime and Airbnb Activity in Chicago

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Introduction:

The City of Chicago is recognized as the third largest city in the United States¹. It has a diverse culture, rich history, and an influence on the economy. It is also one of the major hubs for tourism, with millions of people, each year, visiting places like Millennium Park, the Magnificent Mile, while staying in an ever-growing Airbnb market. However, with the tourism industry, Chicago has faced a big problem with crime, especially homicides. This is an ongoing concern for policymakers, law enforcement, and tourists visiting the area. So, in this report, I will be mapping crime distribution by police district in Chicago to identify high risk areas. To then comparing spatial pattern of homicides, Airbnb listing, Airbnb Prices, and Airbnb room type ratio in Chicago.

Data and Methods:

I started by obtaining data about the police district and crime from the City of Chicago Data Portal² as a zip file. After, I unzipped and imported the police district polygon and crime point shapefile. I then converted the shapefile to NAD 1983 UTM Zone 16N based on my findings from EPSG about UTM of Chicago. I opened the Attributes table with the list and coordinates of all the crimes in Chicago. It included two main types of crime: Human Trafficking (FBI Code 26) and Homicide (FBI Code 1A and 1B). Using the custom shorting tool, I arranged the FBI Code in descending order. Then I removed the crimes that were not listed as Homicide (FBI Code 1A and 1B) from the table. I used the summarize tool to make a summary of the number of Homicides in Primary DE field with the case field as District. I joined the new summary table with the input table of district polygons, input field police district, and output join field police district. I then exported the new feature as its own. I then added a column for the police district area as double to calculate it using the calculate geometry tool in square miles.

I began making a choropleth or graduated color map of homicide density, figure 1, by adding another column to the new feature called Crime Density as double, to calculate the density of homicides per square miles. Using the calculate field, I selected input table as the new joint feature and field name crime density which equaled count Primary DE divided by the district area in miles. I then opened the primary symbology tool of the joint crime layer and selected the graduated color map. I then changed the field to crime density and method to quantile with 6 classes. I made the map light to dark red color scheme. Also, there was a district number that had no data and was not part of Chicago police district so I deleted that from the attribute table.

Then I began making a choropleth map of density of Airbnb reviews, figure 2, by creating a new map within the project and importing the old map. I then obtained, unzipped, and imported

¹ Chicago. "Chicago" Accessed February 5, 2025. <https://www.chicago.gov/city/en/about/facts.html>

² City of Chicago. "City of Chicago Data Portal." Accessed February 5, 2025. <https://data.cityofchicago.org/>.

the review and price data from Inside Airbnb³ website into the map. I followed a similar step like in the beginning in joining the data. First, I summarized this new data by doing a summary of the sum of number of reviews and joined it with the old using the district field. I exported the feature as a new layer. I then added another column to the table for review density and calculated it like previous figure. I then went back to the primary symbology and changed the field to review density, with quantile method and 6 classes. The color scheme of this map was similar to the previous one.

I made a graduated symbol map of the average Airbnb price in each police district, figure 3, using the same methods as before but this time by taking the summary of the Airbnb data for the mean of the statistics field price with the case field district. I then joined the data with the input and output field as district. I then exported the layer as a new feature. After, I went to the primary symbology tab of the new layer and selected graduated symbols. I changed the field to mean price with natural breaks and 6 classes. And the background of the map to very light yellow.

To make a pie chart map, figure 4, that showed the percentage of three Airbnb room types in Chicago by police district, I inserted a new map and copied the map and Airbnb data from the previous one. Then corrected the UTM. I then went open the attribute table for Airbnb data and did a select by attribute where the room type equal shared room and clicked apply. This selected the rows which had the attribute of room type equal to shared room. I then did a count using the summary statistic with field type as room type with the case field district. I then cleared the selection and repeated the process for private room and entire home/apt room type. I then joined all three summaries with the layer for districts of Chicago. I then exported the feature as a new layer so that the data are owned by the layer itself. I then changed the primary symbology to charts and chart type to pie charts. Then I changed the colors of the pie chart to red, blue, and green.

Results:

My findings from choropleth map of homicide density of Chicago by police district, figure 1, show that the middle west side of Chicago has a higher density of homicide per square mile than others. It was shown using light to dark red, where light red meant low homicide density while dark red meant high density of homicide. District 16 and 11 had as high as 11 homicides per square mile but majority of the police districts in Chicago had no more than 3 homicides per square mile.

The choropleth map of Airbnb review density, figure 2, showed more reviews per square miles northeast side of Chicago than others. Figure 2 was also on the scale of light to dark red, where dark red meant high review density and light red meant low review density per square mile. Districts 14, 18, and 19 had high as 2,517 reviews per square miles. Figure 1 and Figure 2 had density regions that were mostly opposite; the highest homicides density area had the lowest review density while the lowest homicides density area had the highest review density.

The graduated symbol map of the average Airbnb price for each police district, figure 3, showed a positive relation with figure 2. The region that had the highest density of Airbnb reviews also had the highest average price. The average price was as high as \$215. The symbols on the

³ Inside Airbnb. "Inside Airbnb." Accessed February 5, 2025. <https://insideairbnb.com/>.

map showed a gradual increase as average prices were higher in that police district. Two of the densest homicide districts in the middle west of Chicago also had higher average prices for an Airbnb. The southern districts had lower reviews and lower average Airbnb prices.

Figure 4 showed a pie chart for each of the police districts in Chicago for the distribution of Airbnb room type. It was red for the entire home/apt, blue for shared room, and green for private room. The distribution of the pie chart was mixed for each police district. Few noticeable ones were district 6 had mostly private rooms, district 7 was mostly entire home/apt, and districts with higher prices had higher shared room type.

Discussion:

In figure 1, the homicide density was more concentrated in the middle west side of Chicago, in police districts 10, 11, 15, and 25. It also include districts in the southern Chicago 3, 6, and 7. The districts with the highest density of homicide was district 15 and 11. It was less concentrated around the northeast side of Chicago in police districts 1, 12, 16, 18, and 19 with district 18 and 16 being the lowest. In figure 2, the Airbnb reviews in Chicago are more concentrated around the northeast side of Chicago in districts 14, 18, 19, 20 and 24 with district 14, 18, 19 being the highest. While the south and southwestern regions have the lowest. The prices in the regions with higher Airbnb rating also had a higher average price, shown in figure 3. The less average prices were regions with less Airbnb review density. Figure 4 showed a mixture of three types of room with the shared room being the majority in regions with higher average Airbnb prices.

From my understanding, the northeast side of Chicago is more popular because it has a mix of attractions that attract more tourists. This results in higher demand for a location to stay in and that drives the average price for Airbnb to be higher. That side is also where Lake Michigan is located which allows for scenic views and easy access to the waterfront. The higher average price for Airbnb in that region also causes more share room type to lower the cost of staying there. The homicide density in the middle west of Chicago might be high due to the lack of economic opportunities, more gang activity, and could also be because of underfunded law enforcement.

Conclusion:

There is a lot happening in different police districts in the city of Chicago, with high homicide density per square mile in the middle west, high Airbnb reviews and Airbnb prices in the east. The correlation I saw from the four figures is that there are fewer visitors living at an Airbnb where there is high homicide density. It is not clear as to why there is more homicide happening in the middle west side of Chicago. However, it could be better understood if more data was provided on how many police officers there are in each district. Additionally, the census data for each police district would have been beneficial in deciding the density of homicide based on the number of people living in the area. It could be that more people living in that area and so more homicide per square mile. But it is clear that Airbnb prices at different locations caused major people in the higher average cost area to select share room type to lower their cost of staying there a few nights.

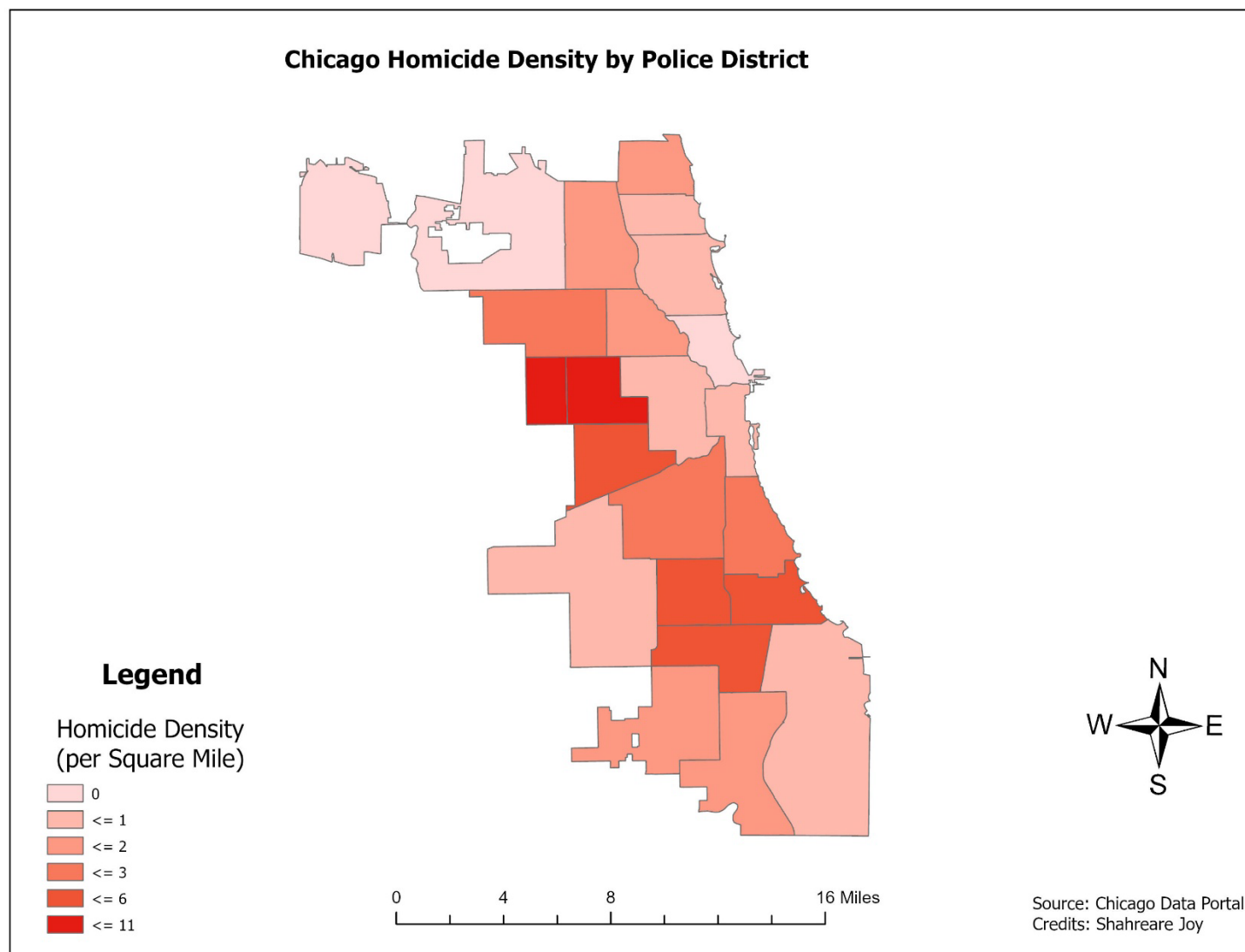


Figure 1: A choropleth map showing the density of homicides (per square mile) in Chicago by its police district.

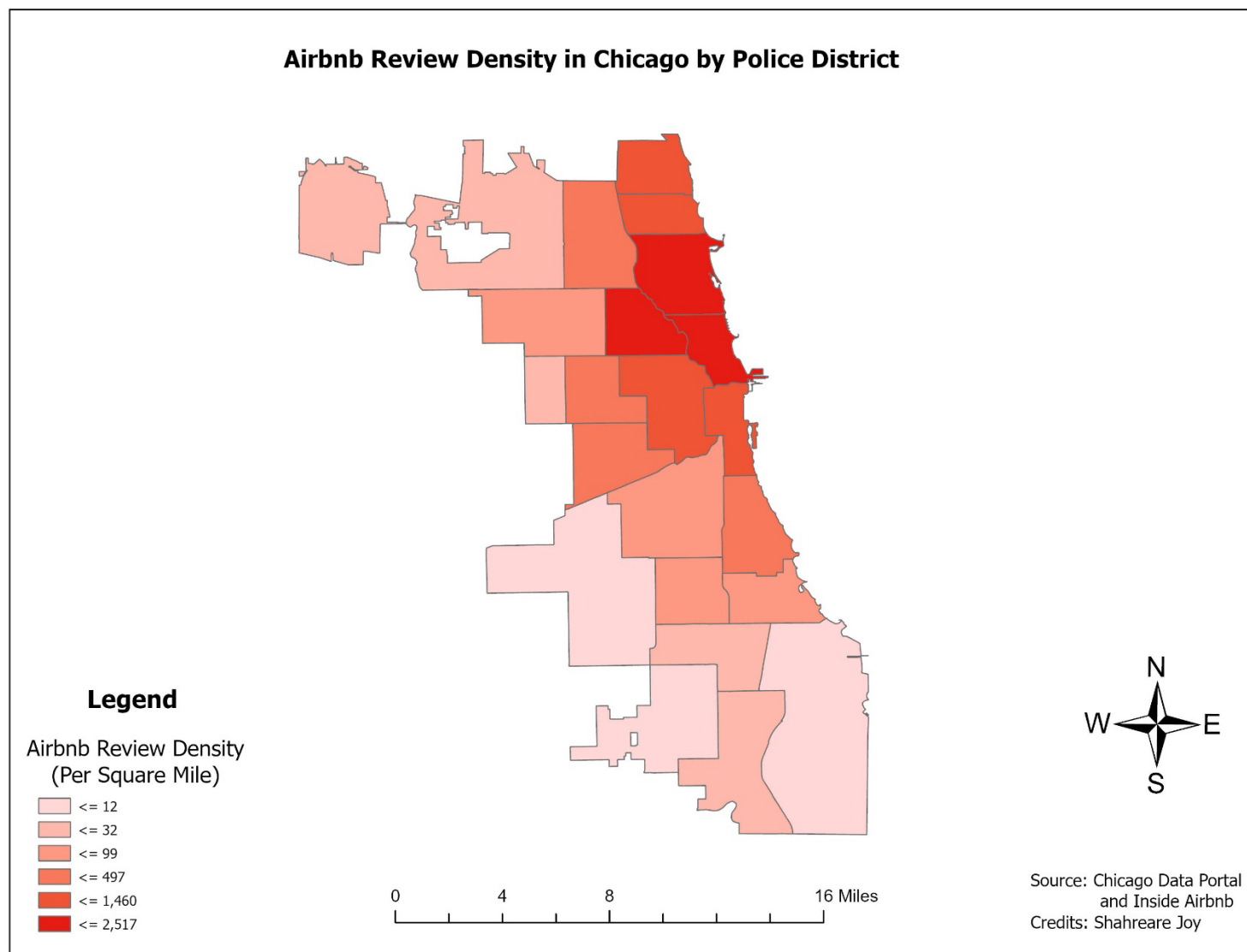


Figure 2: A choropleth map showing the density of Airbnb review (per square mile) in Chicago by its police districts.

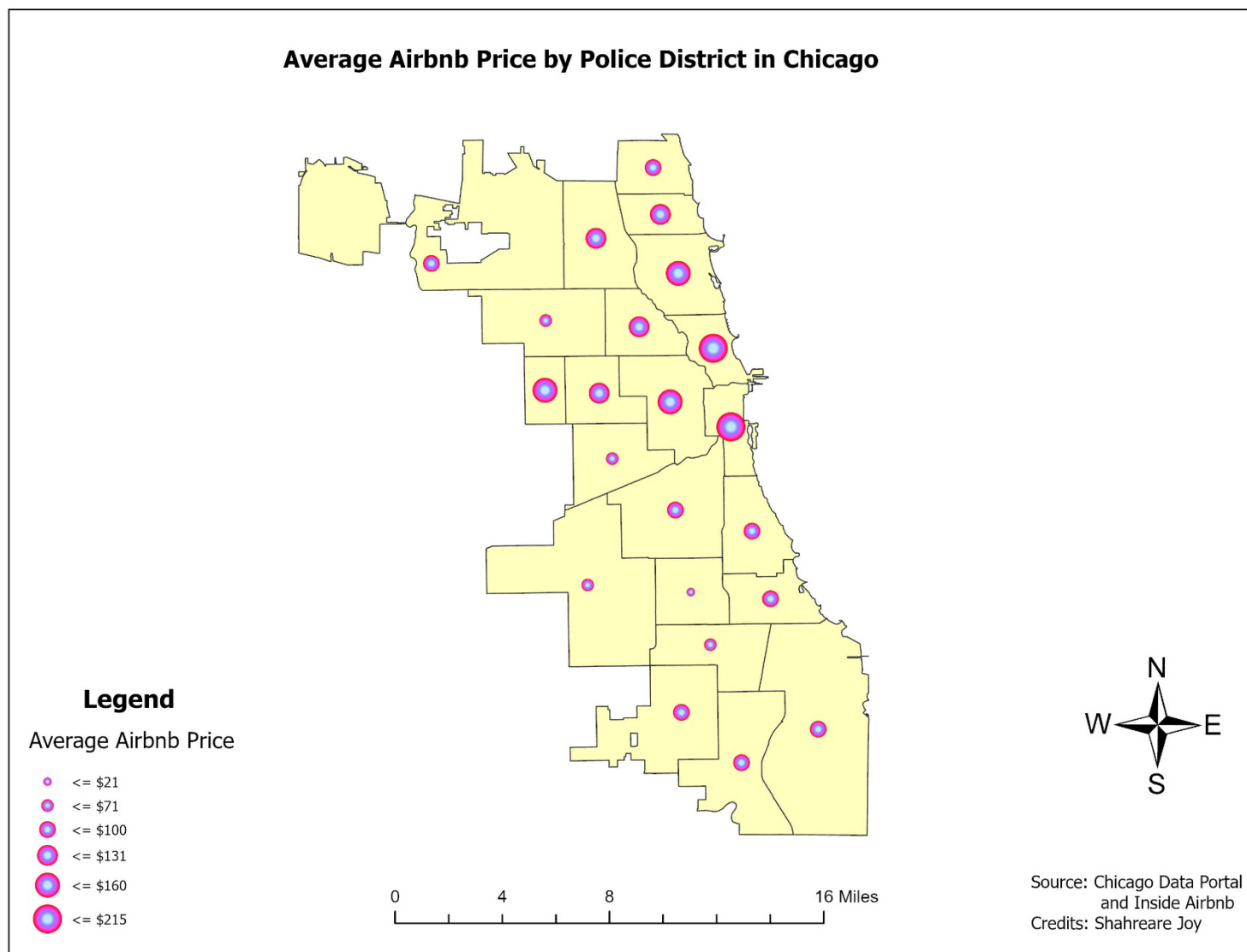


Figure 3: A graduated symbol map showing the average Airbnb price in each of the police districts in Chicago.

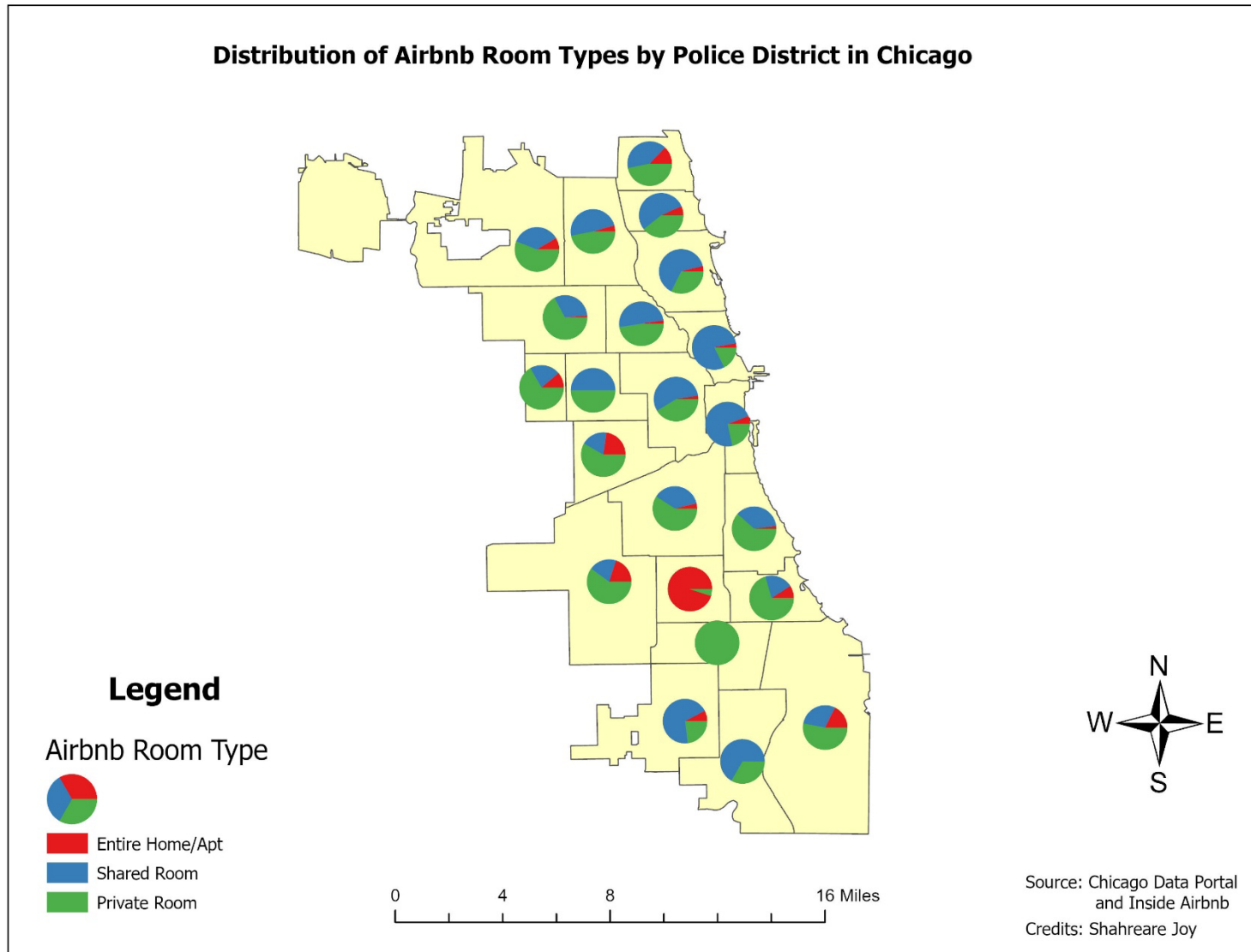


Figure 4: Pie chart graph showing the percentage of three Airbnb room types in Chicago by its police district.